CS & IT ENGINEERING

DISCRETE MATHS
SET THEORY

Lecture No. 11

By- SATISH YADAV SIR

TOPICS TO BE COVERED

01 Basics of relations

02 Types of relations

03 Number Of relations

Relations

$A = \{1, 2, 3\}$

$\left\{ \begin{array}{l} (P(A), \subseteq) \text{ poset} \\ (\text{set}, \text{Relation}) \end{array} \right.$

$n = 3$

$e = n \cdot 2^{n-1}$

$= 3 \cdot 2^{3-1}$

$= 3 \cdot 4 = \underline{\underline{12}}$

$n = 5$

$\underline{R}$: $A \, R \, A$ $\quad A \subseteq A$

(R: $\subseteq$)

$\underline{\text{Anti}}$: $ARB \wedge BRA \longrightarrow A = B$

$A \subseteq B \wedge B \subseteq A \longrightarrow A = B$ ✓

$\{1\} \subseteq \{1, 2\} \wedge \{1, 2\} \subseteq \{1\} \longrightarrow$

* $\underline{T}$: $\underline{A \subseteq B} \wedge \underline{B \subseteq C} \longrightarrow \underline{A \subseteq C}$

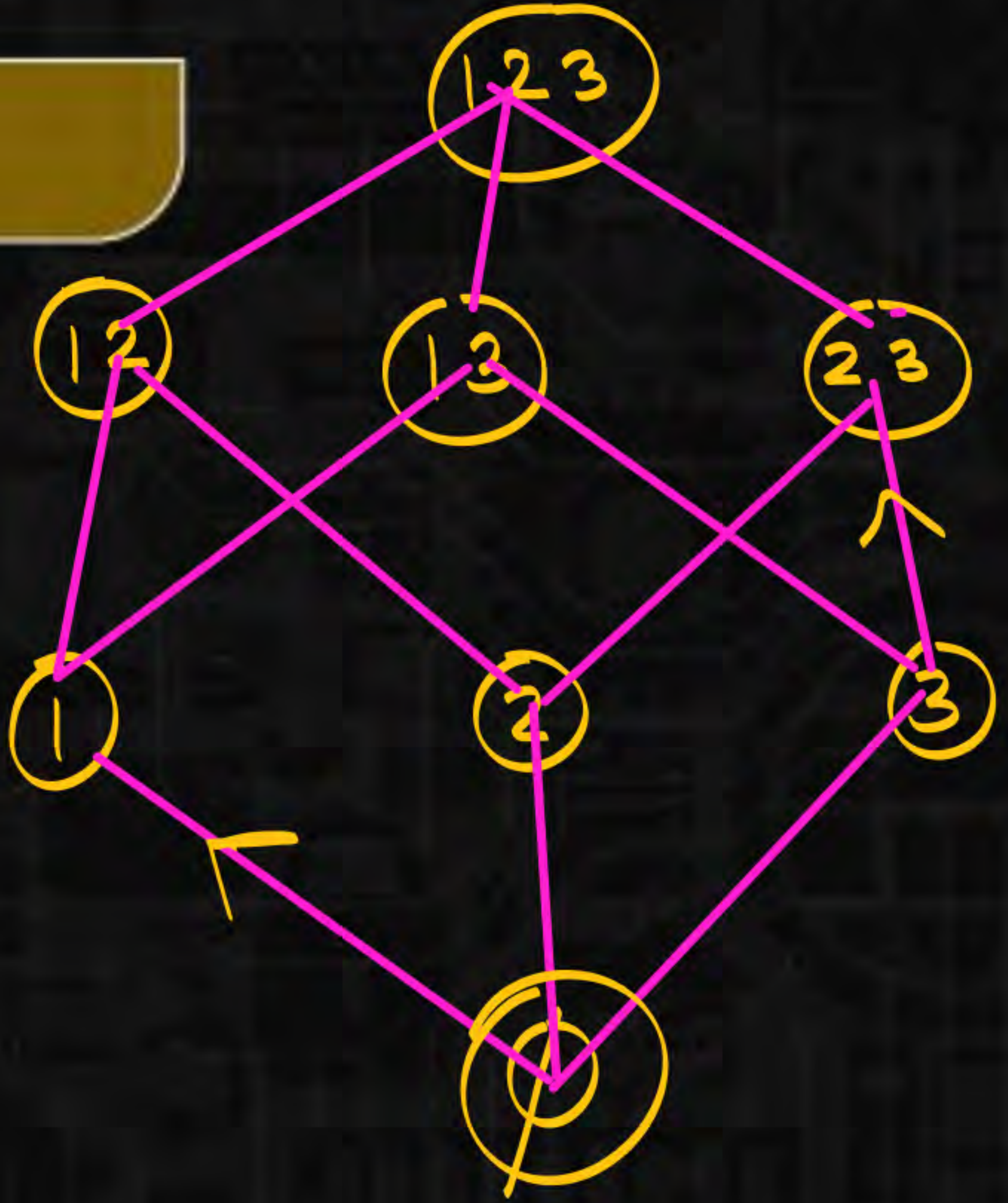
Relations

$A = \{1, 2, 3\}$

$(P(A), \subseteq)$

$P(A)$ → set

$\subseteq$ → Relation

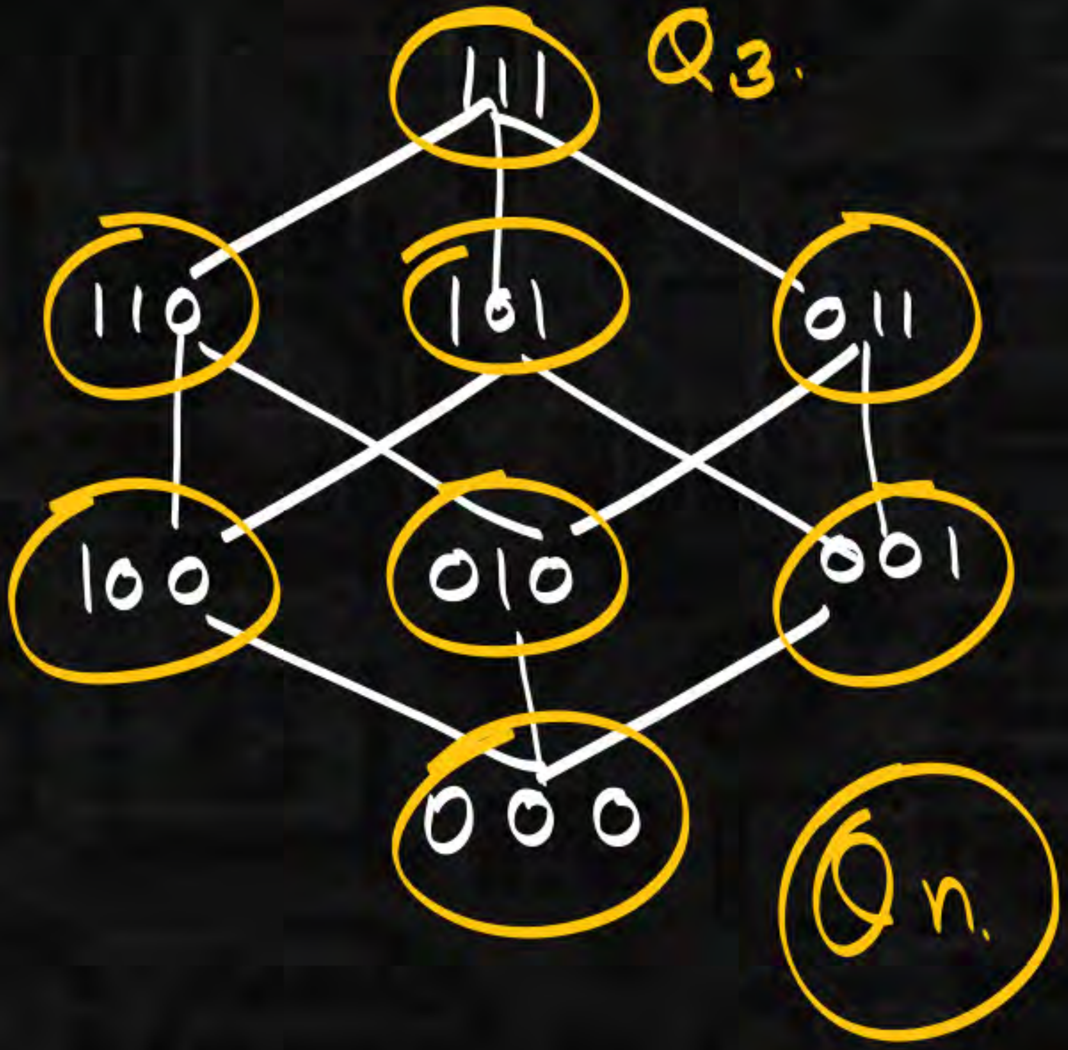


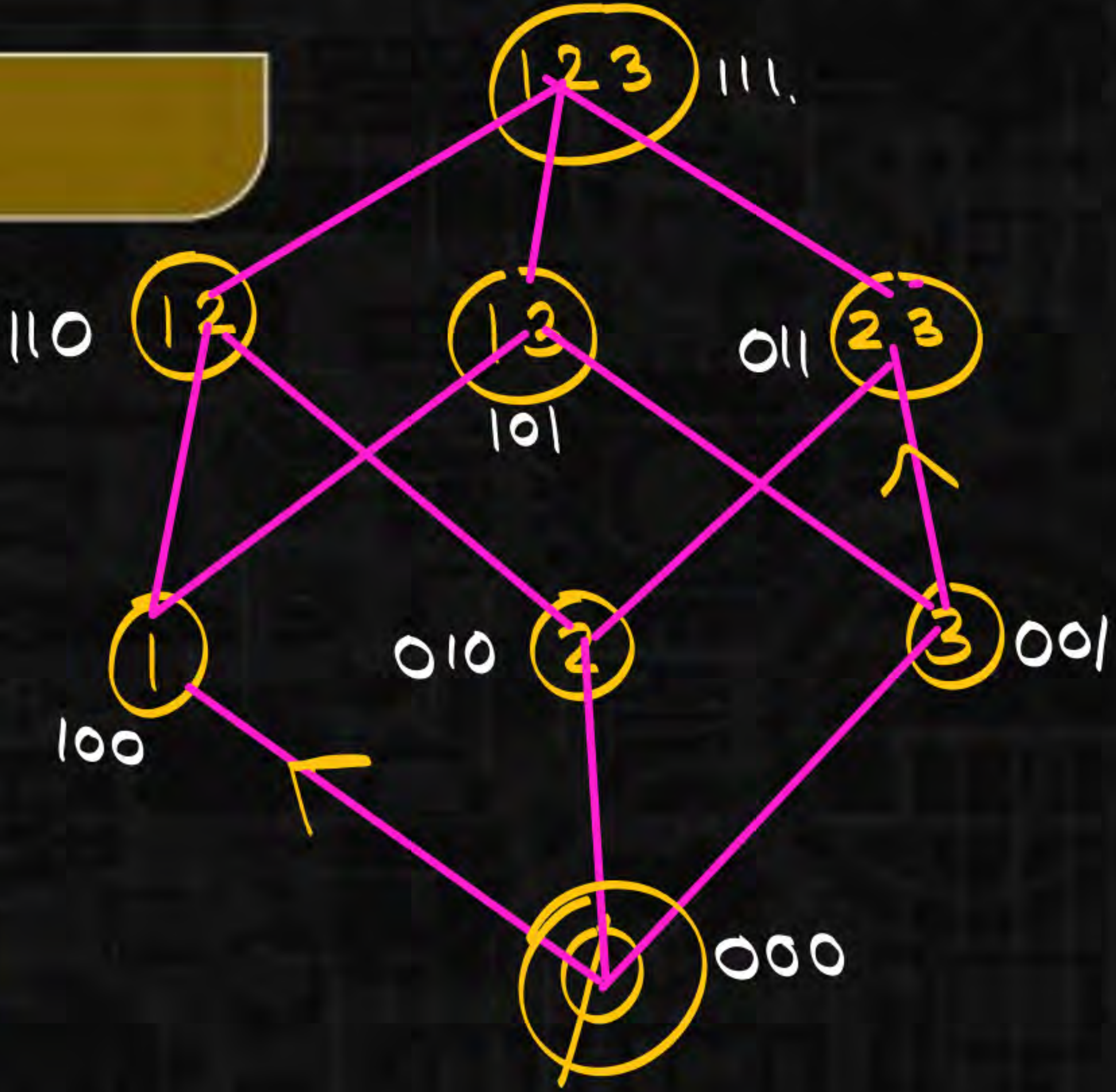
$\{1\,2\} \not\subseteq \{1\,3\}$

$\{1\,3\} \not\subseteq \{1\,2\}$

$(\{\phi, \{1\}, \{2\},$

Relations





1	2	3	
0	0	0	$\rightarrow \phi$
1	0	0	$\rightarrow \{1\}$

Relations

Qn. Degree of each vertex $= n$.

Total vertices $= 2^n$.

$$\boxed{e = n \cdot 2^{n-1}}$$

$$\Sigma d(v_i) = 2e.$$

$$n \cdot 2^n = 2e$$

$A = \{1, 2, 3, 4, 5\}$

$(P(A), \subseteq)$

$n = 5$

$$e = n \cdot 2^{n-1}$$
$$= 5 \cdot 2^4 = 5 \cdot 16 = 80$$

Relations

Assumption:

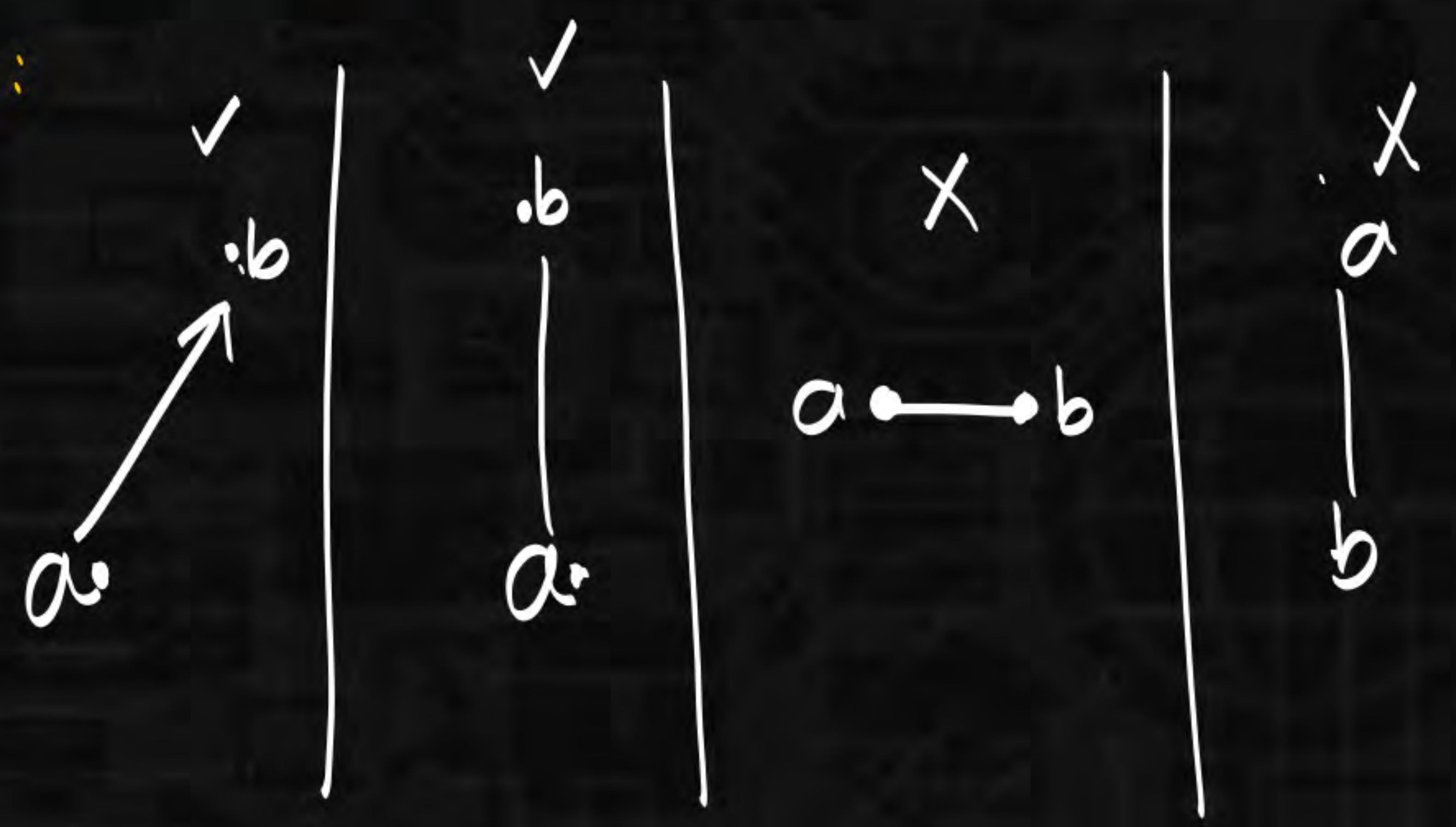
$(a, b) \in R$

OR

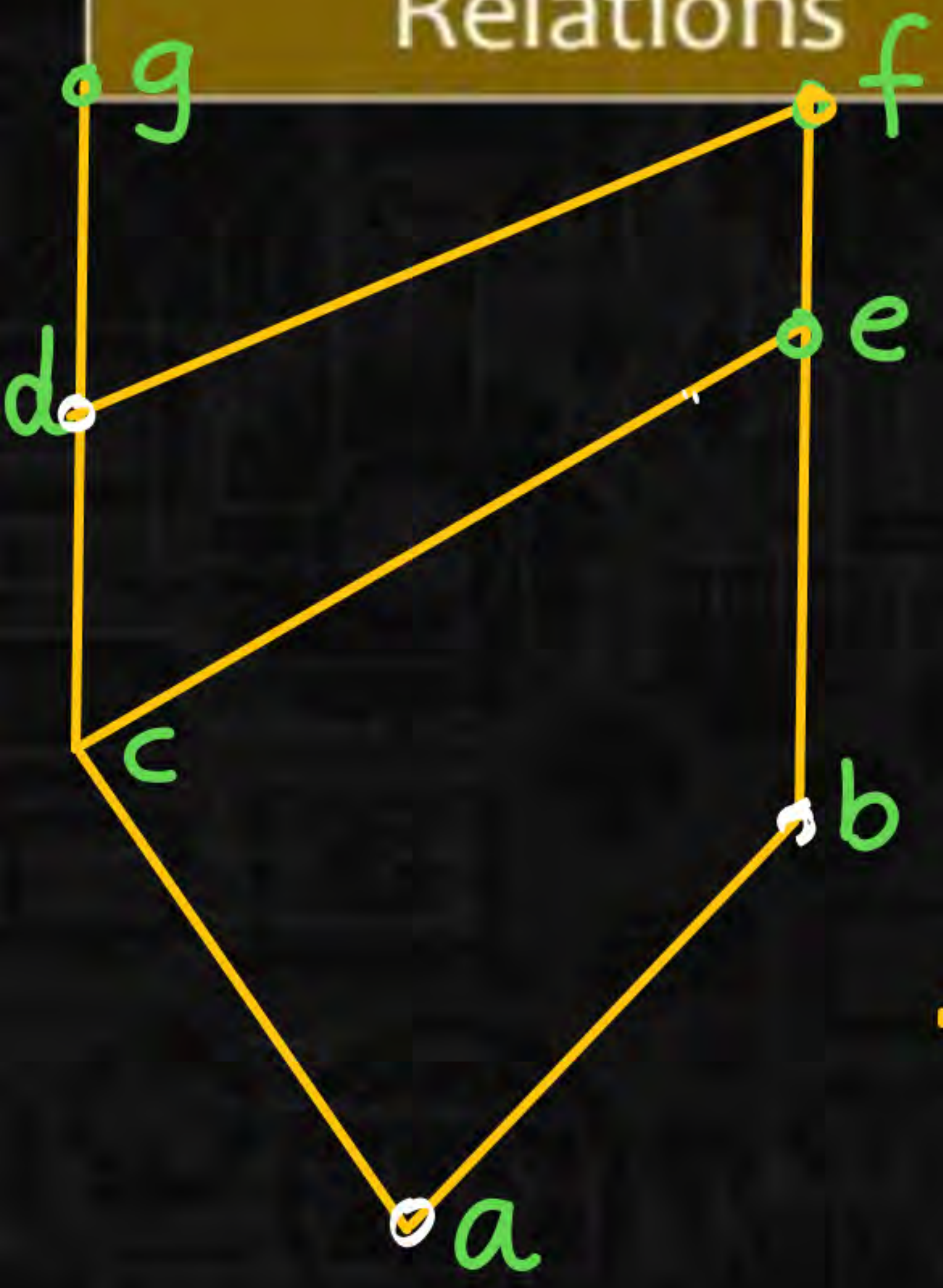
aRb

$a \leq b$

Hasse:



Relations



(A, R) poset.

Greatest element (GE)

x is called greatest element

all elements $\in A \leq x \in A$.

Check for f:

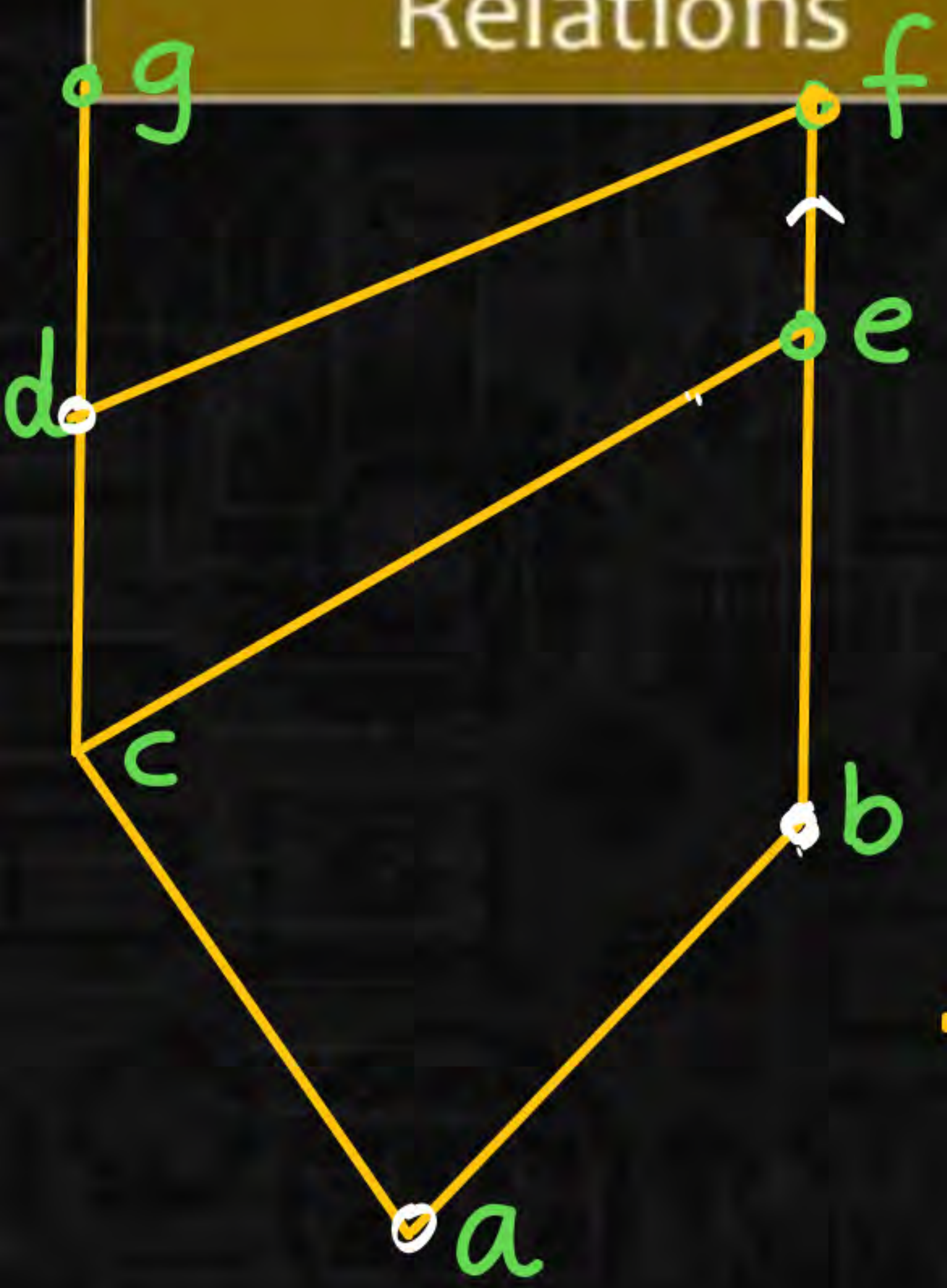
abcdefg $\leq f$

$a \leq f$ ✓ $\quad c \leq f$ ✓ $\quad e \leq f$ ✓

$b \leq f$ ✓ $\quad d \leq f \quad g \leq f$ (✗)

f is not greatest element.

Relations



(A, R) poset.

Greatest element (GE)

x is called greatest element

all elements $\in A \leq x \in A$.

for g:

$a \leq g$ ✓

abcdefg $\leq$ g. $\quad b \leq g$ (X) ...

Relations

It is not necessary such that every Hasse diagram should contains greatest element

Thm: if G.E exist then it will be unique.

G.E x_1 G.E x_2.

all elements $\leq x_1$ all elements $\leq x_2$

$\therefore x_2 \leq x_1 \quad \wedge \quad \therefore x_1 \leq x_2 \longrightarrow x_1 = x_2.$

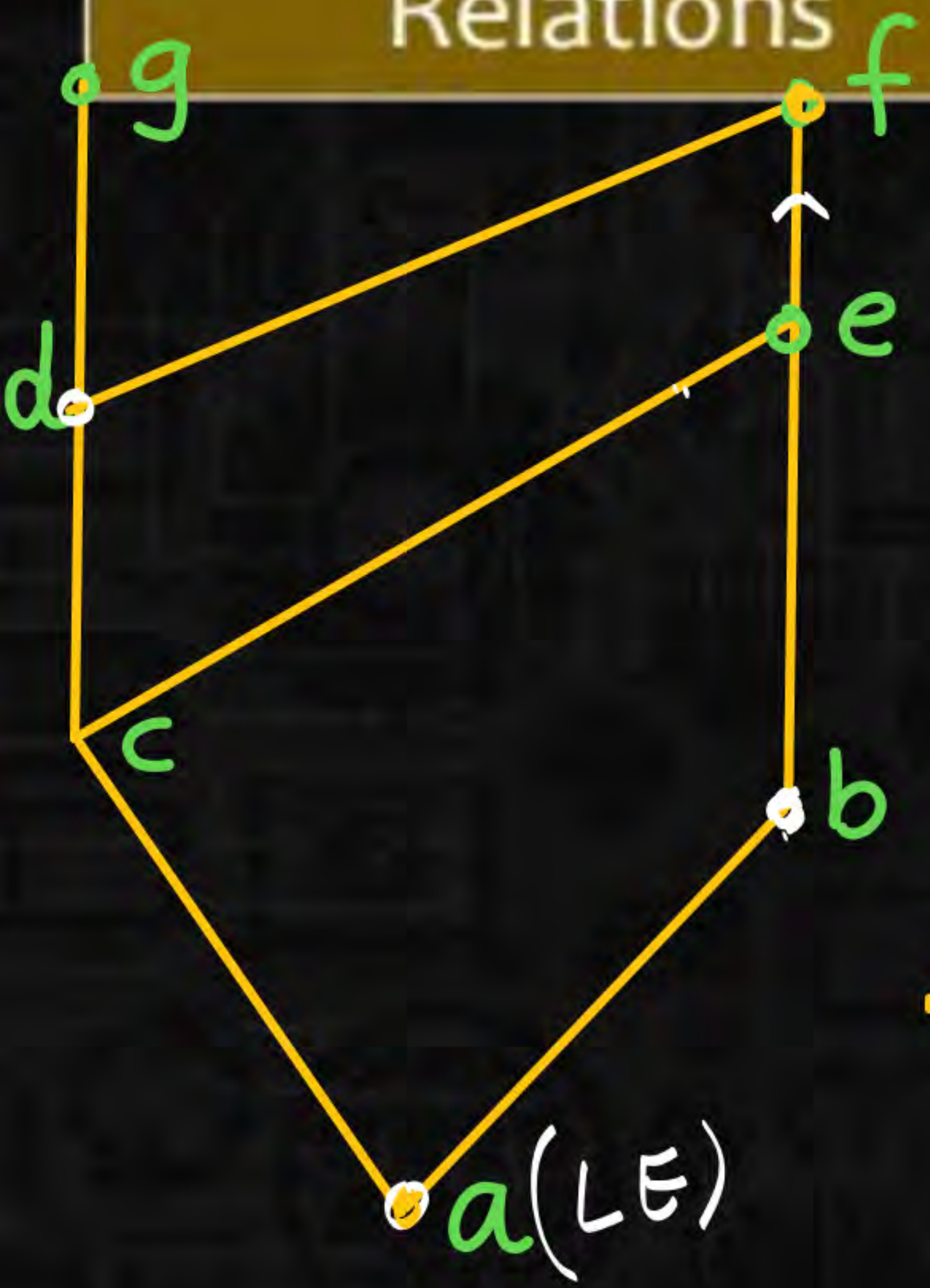
Relations

$(D_{12}, |)$

all elements ≤ 12

$$\begin{cases} 1 \leq 12 \\ 2 \leq 12 \\ 3 \leq 12 \\ 4 \leq 12 \\ 6 \leq 12 \\ 12 \leq 12 \text{ (Ref)} \end{cases}$$

Relations



(A, R) poset.

least element (LE)

x is called least element

$x \in A \leq$ all elements $\in A$.

$a \leq$ all elements.

$a \leq abcdefg$.

$a \leq a$ ✓ $a \leq b$ ✓ $a \leq c$ ✓ $a \leq d$ ✓ $a \leq e$ ✓ $a \leq f$ ✓ $a \leq g$ ✓

Relations

Thm: if LE exist then it will be unique.

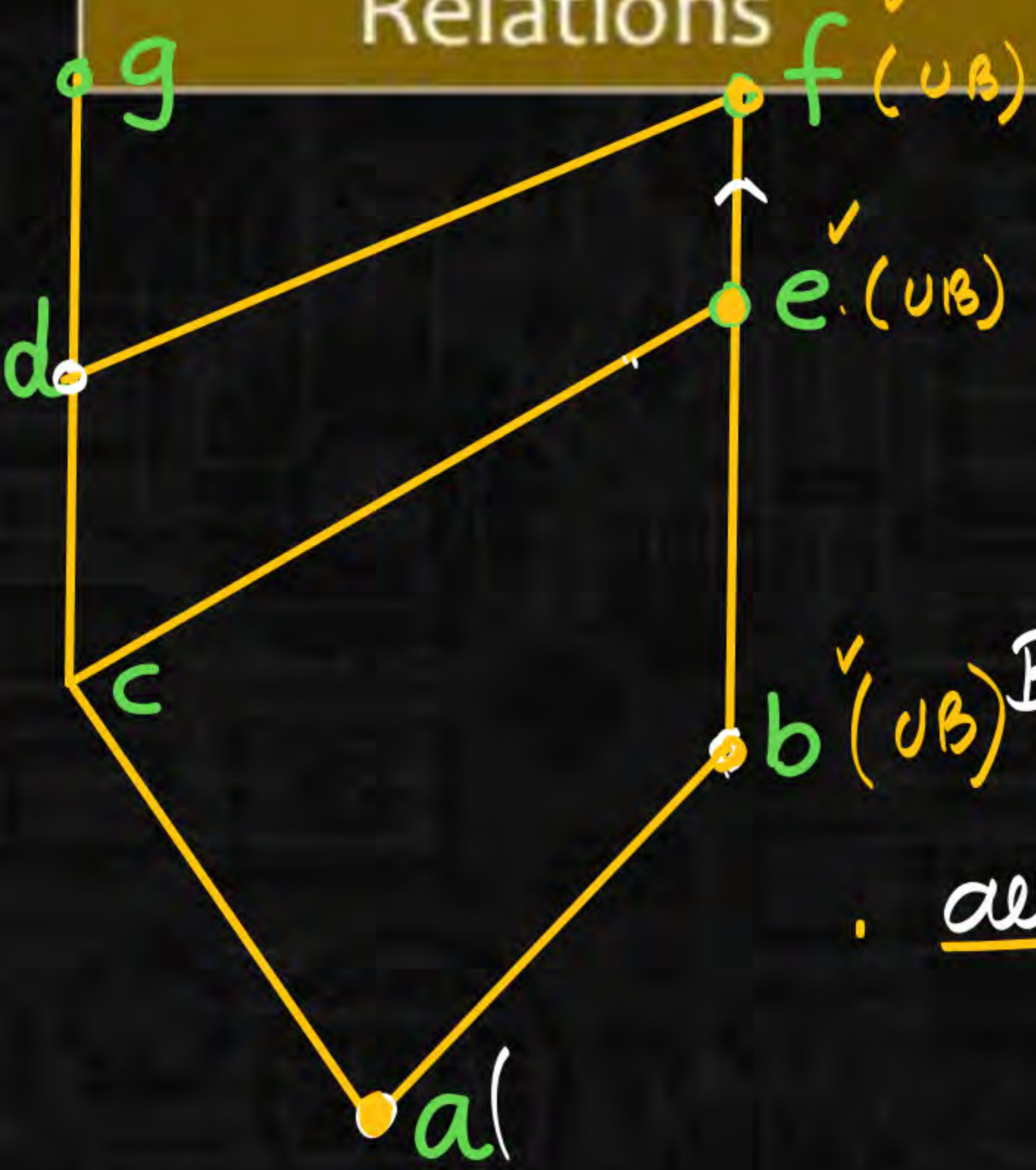
I.

II.

III.

IV.

Relations



(A, R) poset

$B \subseteq A$

Upper bound (UB)

x is called UB of B

all elements of $B \leq x \in A$

$B = \{a\ b\}$

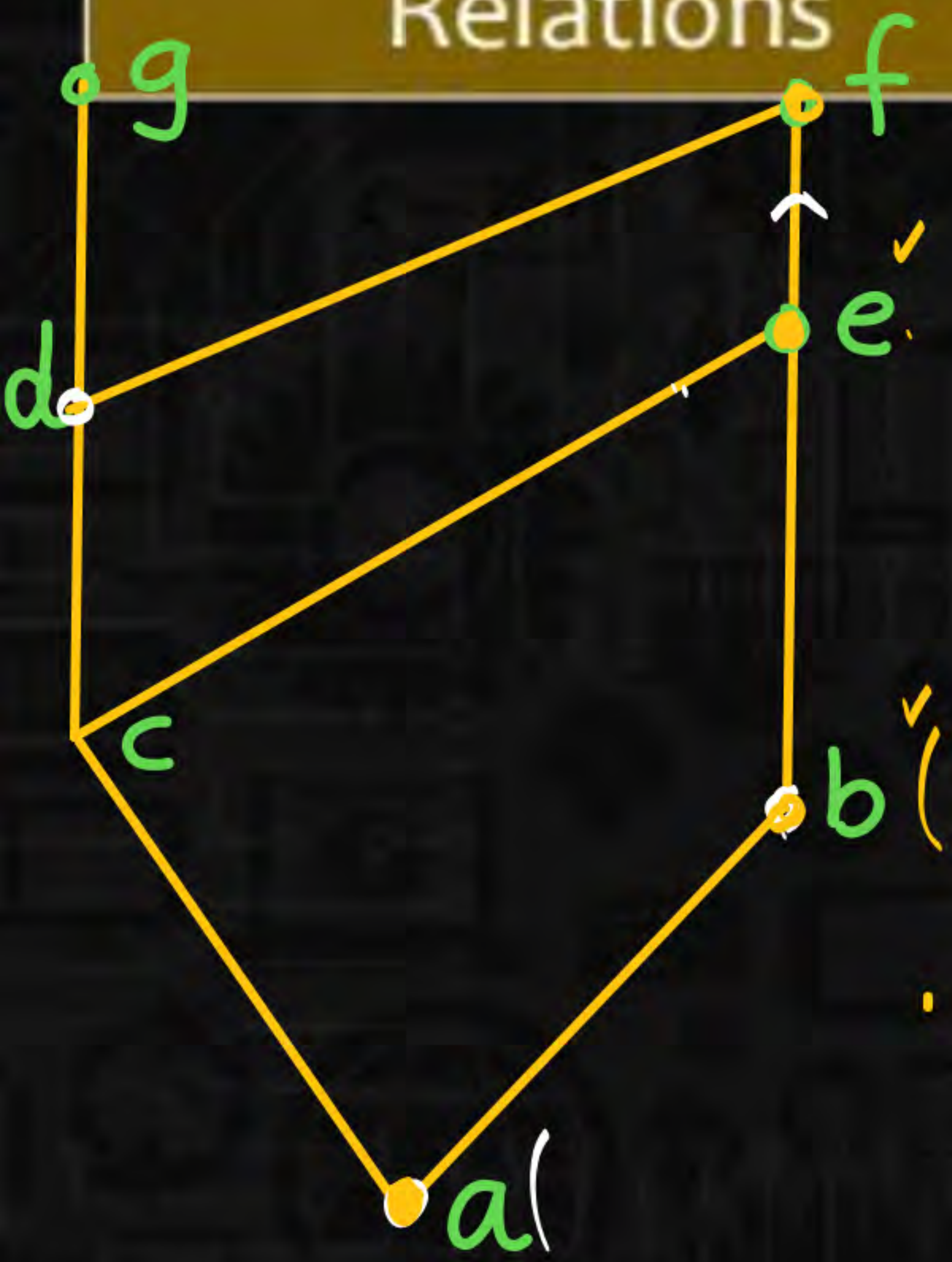
all elements of $B \leq x \in A$

$ab \leq e$ | $ab \leq f$ | $ab \leq b$

$a \leq b$ ✓

$b \leq b$ ✓

Relations



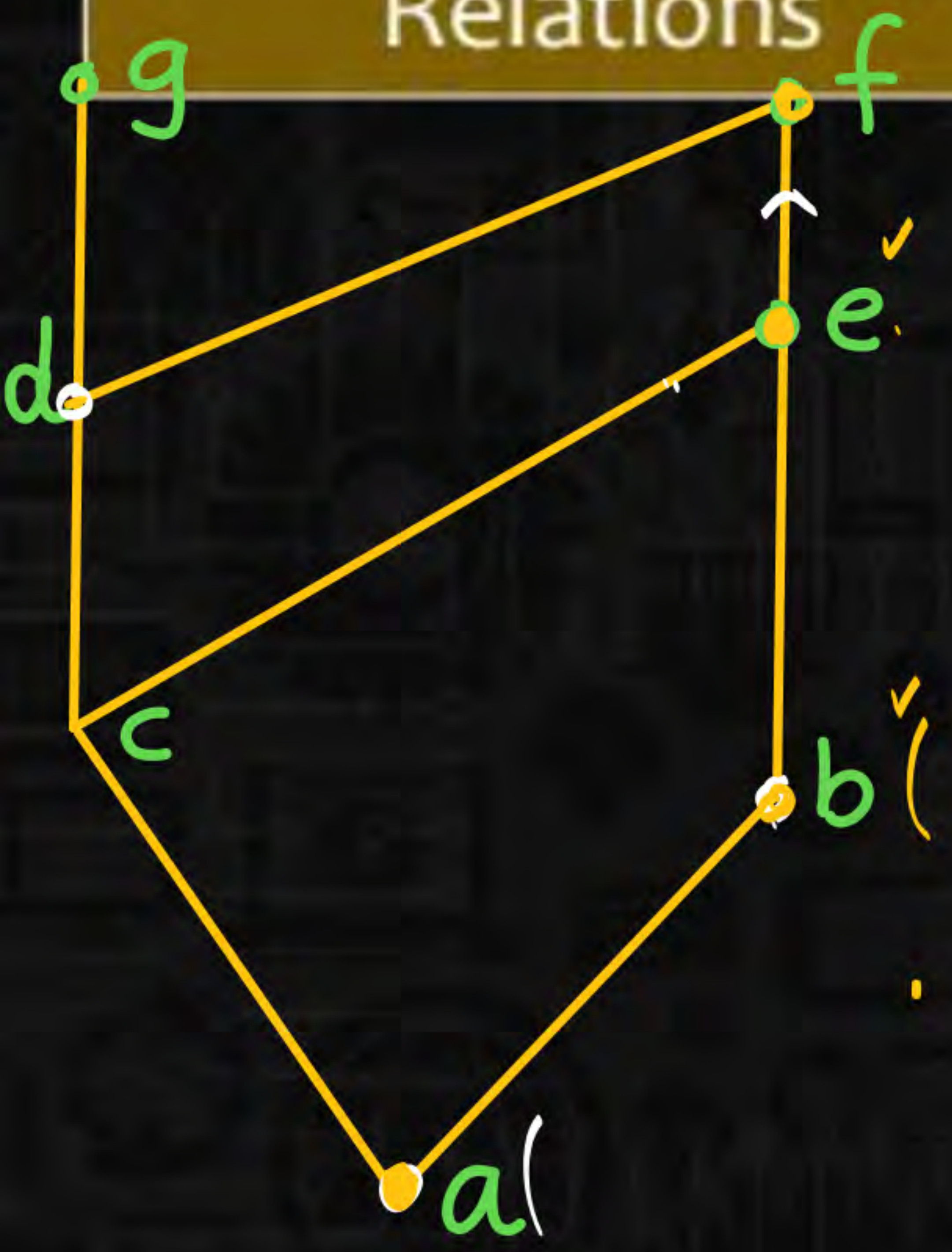
$B = \{c, b\}$

e, f are UB of $\{c, b\}$.

$cb \le e$	$cb \le f$
$c \le e$ ✓	$c \le f$ ✓
$b \le e$ ✓	$b \le f$ ✓

$B_1 = \{d, b\}$

Relations



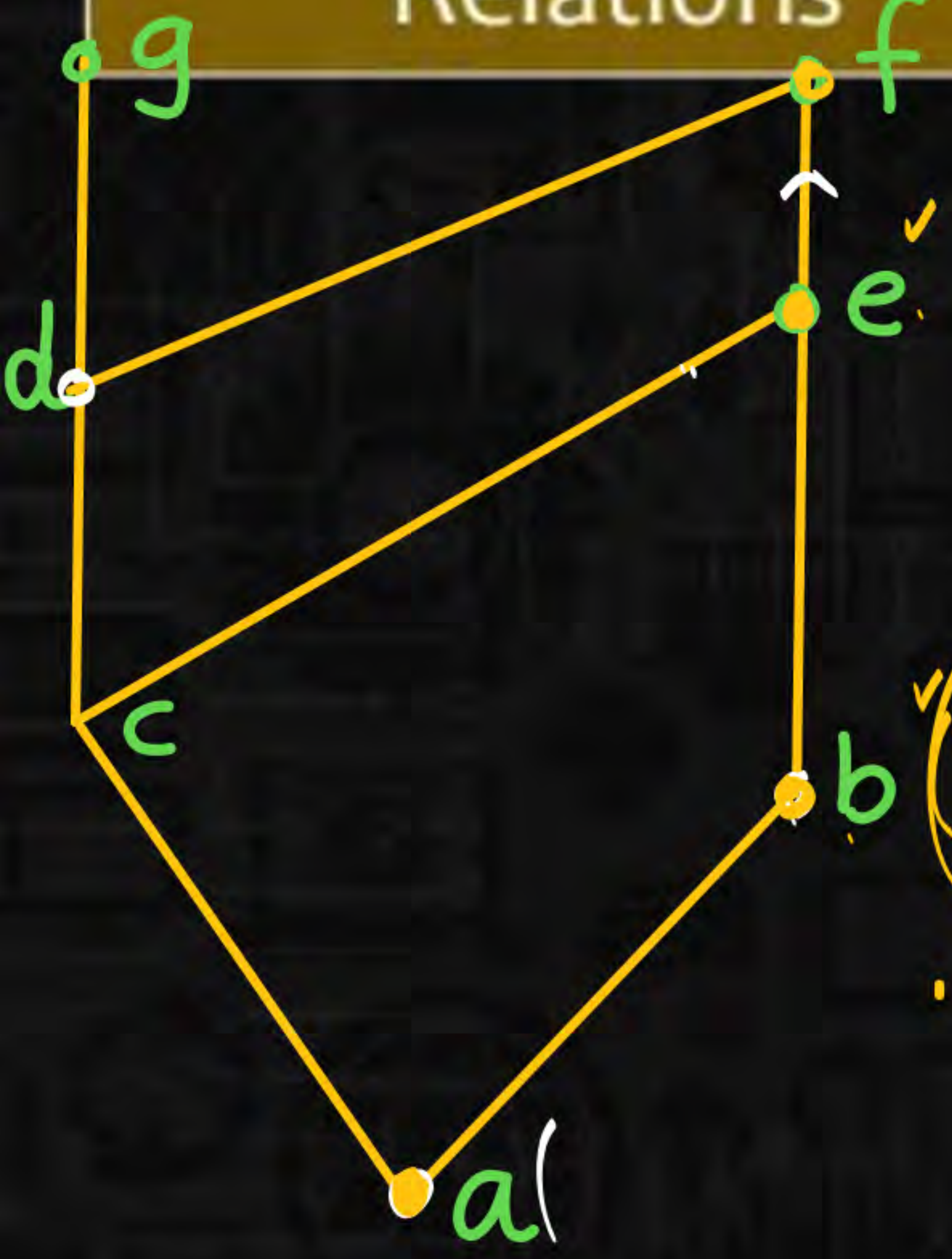
UB:

LUB : (~~least~~ upper bound)

$\boxed{\text{all elements of } B \leq x \in A.}$

LUB $\leq$ all UB's of B.

Relations



$B = \{a, b\}$

UB's of $B = \{b, e, f\}$

LUB $\le$ bef

(L.U.B) check for b

$b \le bef$

$b \le b$ ✓

$b \le e$ ✓

$b \le f$ ✓

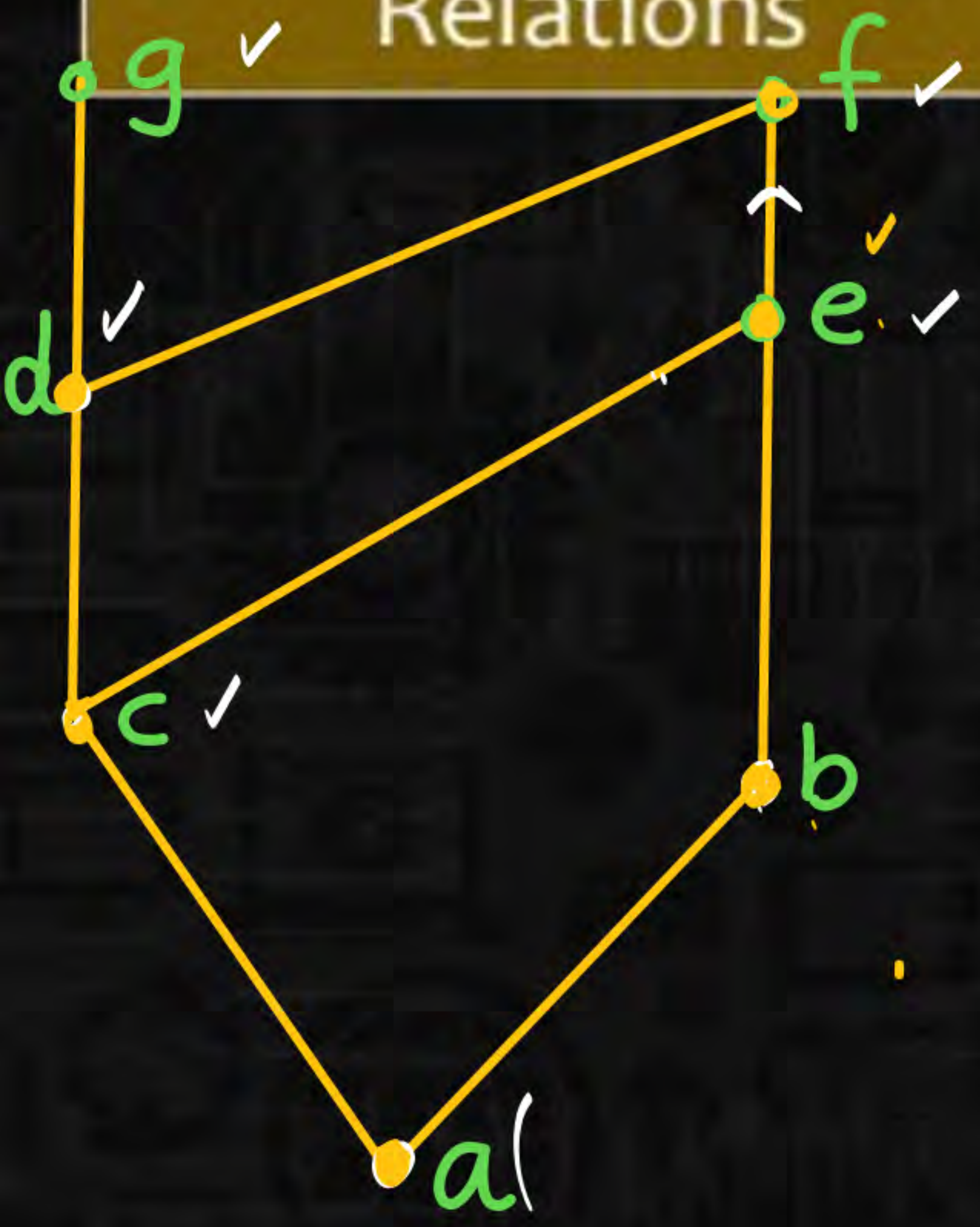
for e

$e \le bef$

for f

$f \le bef$

Relations



bg ≤

$B = \{a, c\}$

UB's of $\{a, c\} = \{c, e, d, f, g\}$.

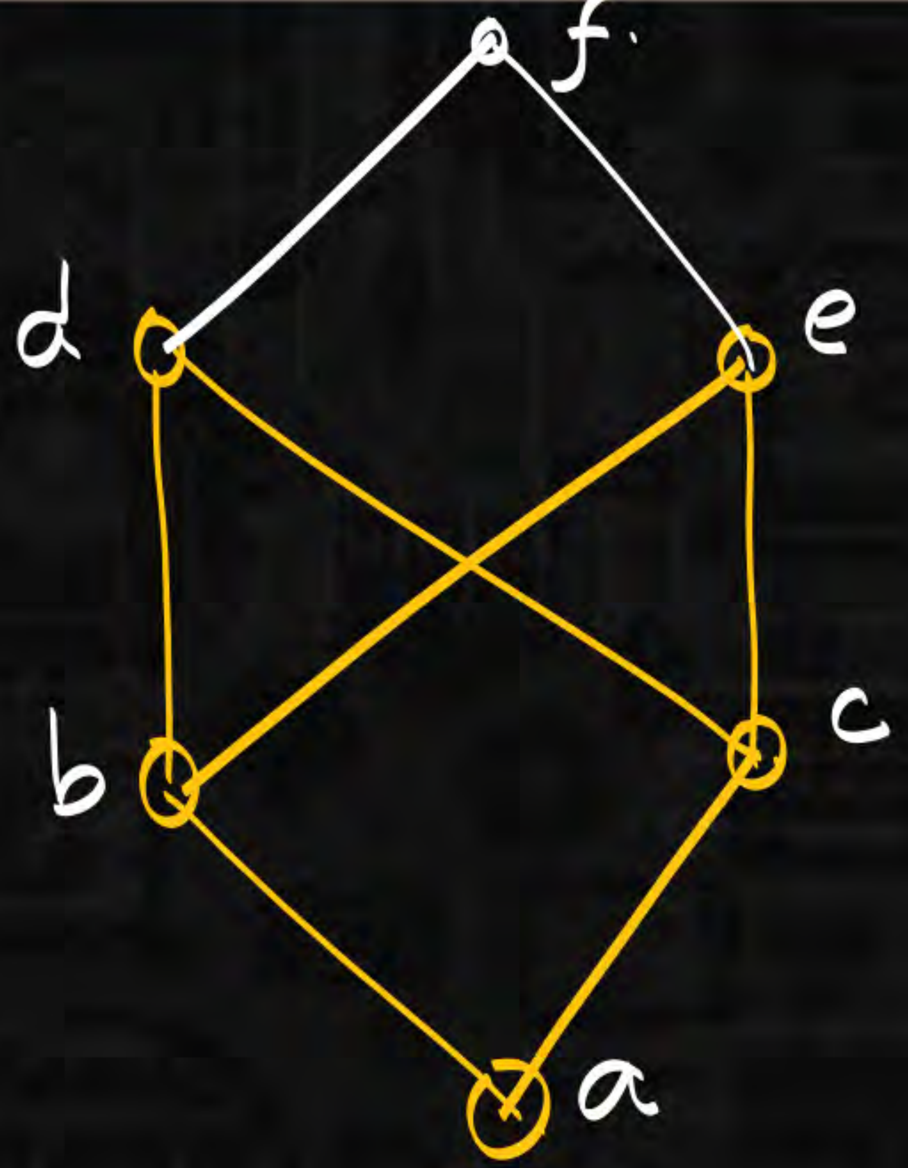
$\underline{\text{LUB of } \{a c\}} = \{c\}$

$c \leq c\,e\,d\,f\,g$.

$lub(d, e) = f$

$lub(b, g) =$ NA.

Relations



$lub(b,c) = N.A.$

UB's of $\{b,c\} = \{d, e, f\}$

$lub \le$ all UB's of B

$\boxed{\text{If LUB exist it will be unique.}}$

$d \le def$	$e \le def$	$f \le def$
$d \le d$ ✓	$e \le d$ ✗	$f \le d$ ✗
$d \le e$ ✗		

Relations

(A, R) poset

$B \subseteq A$.

LB: (lower bound)

x is LB of B.

$x \in A \leq$ all elements of B.

$B = \{d, e\}$

$c \leq de$ | $a \leq ce$

$\{a, c\}$ is LB of $\{d, e\}$

Relations

(A, R) poset

glb : (~~greatest~~ lower bound) $B \subseteq A$.

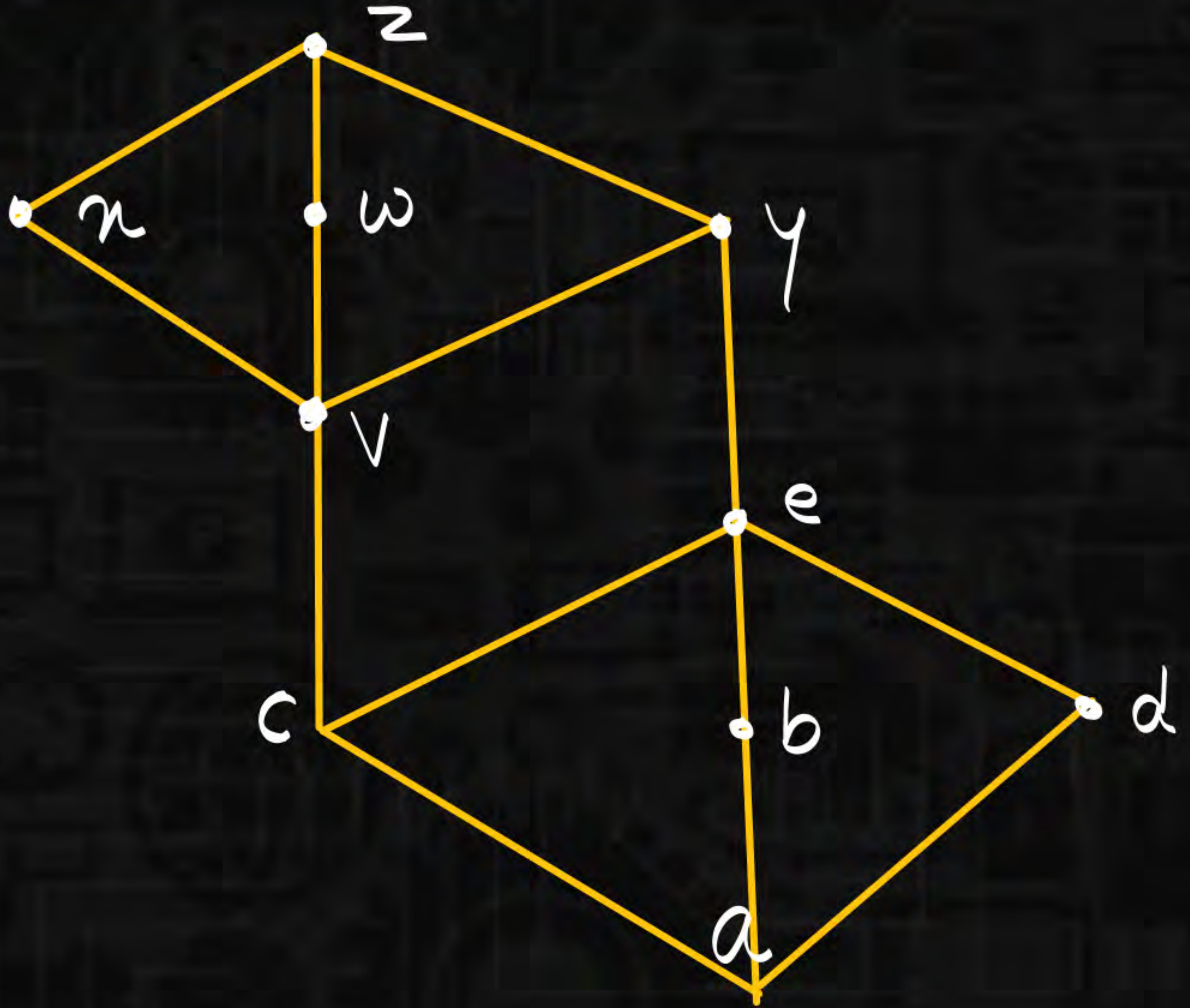
$B = \{g, f\}$

lb's of $\{g, f\} = \{d, c, a\}$

$d \ge c$, $a \le d$ | $d \ge c$, $a \le c$ | $d \ge c$, $a \le a$.

d is glb of $\{g, f\}$

Relations



Hw

glb(bc)

glb(bw)

glb(ex)

lub(cb)

lub(dx)

lub(ce)

lub(av)


THANK YOU GW
SOLDIERS !